

Lab: Cassandra Table Operations



Objectives

After completing this lab you will be able to:

- Create a table in a keyspace by defining a column name and data type
- Extract the details of a table with the DESCRIBE command
- Alter a table by adding columns
- Drop a table by removing it from the keyspace

Exercise 1 - Getting the environment ready

Open the Cassandra database page by clicking the button below.

 [Open Cassandra Page in IDE](#)

On that page, click the [Create](#) button to create a Cassandra database.

After creation has finished, click the [Cassandra CLI](#) button (below the [Create](#) button from the previous step) to connect to the database using the cqlsh CLI.

1. Problem:

Create a keyspace named **training** using SimpleStrategy and replication factor of 3.

▼ Click here for Hint

use `CREATE KEYSPACE` command with appropriate options

▼ Click here for Solution

On the cqlsh run the below command.

```
1 CREATE KEYSPACE training
2 WITH replication = {'class': 'SimpleStrategy', 'replication_factor' : 3};
```



Exercise 2 - Create a table

The below command creates a table named **movies**, in the **training** keyspace.

The **movies** table has three columns:

- 'movie_id' is an integer and is the primary key.
- 'movie_name' is a text column.
- 'year_of_release' is an integer.

On the cassandra client run the below command.

```
1 use training;
2 CREATE TABLE movies(
3   movie_id int PRIMARY KEY,
4   movie_name text,
5   year_of_release int
6 );
```

Verify that the table got created by listing all tables.

```
1 describe tables;
```

```
root@cqlsh:system> CREATE KEYSPACE training
... WITH replication = {'class':'SimpleStrategy', 'replication_factor' : 3};
```

Warnings :

Your replication factor 3 for keyspace training is higher than the number of nodes 1

```
root@cqlsh:system> use training;
root@cqlsh:training> CREATE TABLE movies(
...   movie_id int PRIMARY KEY,
...   movie_name text,
...   year_of_release int
... );
```

```
root@cqlsh:training> describe tables;
```

movies

```
root@cqlsh:training> []
```

Exercise 3 - Describe a table

In the previous exercise you created a table named **movies**.

Let us print more details of it using the describe command.

Describe the table.

```
1 describe movies
```

```

root@cqlsh:training> describe movies

CREATE TABLE training.movies (
  movie_id int PRIMARY KEY,
  movie_name text,
  year_of_release int
) WITH additional_write_policy = '99p'
  AND allow_auto_snapshot = true
  AND bloom_filter_fp_chance = 0.01
  AND caching = {'keys': 'ALL', 'rows_per_partition': 'NONE'}
  AND cdc = false
  AND comment = ''
  AND compaction = {'class': 'org.apache.cassandra.db.compaction.SizeTieredCompactionStrategy', 'max_threshold': '32', 'min_threshold': '4'}
  AND compression = {'chunk_length_in_kb': '16', 'class': 'org.apache.cassandra.io.compress.LZ4Compressor'}
  AND memtable = 'default'
  AND crc_check_chance = 1.0
  AND default_time_to_live = 0
  AND extensions = {}
  AND gc_grace_seconds = 864000
  AND incremental_backups = true
  AND max_index_interval = 2048
  AND memtable_flush_period_in_ms = 0
  AND min_index_interval = 128
  AND read_repair = 'BLOCKING'
  AND speculative_retry = '99p';
root@cqlsh:training>

```

Exercise 4 - Alter a table

In a previous exercise you created a table named **movies**.
 Let us modify it by adding a column named 'genre' which is of type 'text'.

Alter the table.

```

1 ALTER TABLE movies
2 ADD genre text;

```

Verify the changes using the below command

```

1 describe movies;

```

Exercise 5 - Drop a table

To drop the movies table run the below command.

```

1 drop table movies;

```

Verify using the below command. You should get an error.

```

1 describe movies;

```

Practice exercises

1. Problem:

Create a table named *books* with 3 columns; *'book_id'* which is the primary key and of integer type, *'author'* which is of type text and *'title'* which is of type text.

▼ Click here for Hint

use the *'create table'* command

▼ Click here for Solution

On the cqlsh run the below command.

```
1 CREATE TABLE books (  
2   book_id int PRIMARY KEY,  
3   author text,  
4   title text  
5 );
```



2. Problem:

Add a column *'date_of_publication'* which is of date type.

▼ Click here for Hint

use the *'alter table'* command

▼ Click here for Solution

On the cqlsh run the below command.

```
1 ALTER TABLE books  
2 add date_of_publication date;
```



3. Problem:

Add a *'column'* price which is of type decimal.

▼ Click here for Hint

use the *'alter table'* command

▼ Click here for Solution

On the cqlsh run the below command.

```
1 ALTER TABLE books  
2 add price decimal;
```



4. Problem:

Drop the 'column' price from the **books** table.

▼ Click here for Hint

use the 'alter table' command with drop option

▼ Click here for Solution

On the cqlsh run the below command.

```
1 ALTER TABLE books
2 drop price;
```

5. Problem:

Drop the **books** table.

▼ Click here for Hint

use the 'drop table' command

▼ Click here for Solution

On the cqlsh run the below command.

```
1 DROP TABLE books;
```

6. Problem:

Drop the **training** keyspace.

► Click here for Hint

▼ Click here for Solution

On the cqlsh run the below command.

```
1 drop keyspace training;
```

7. Problem:

Disconnect from the cassandra server.

▼ Click here for Hint

Refer to the [Using the CQL Shell \(cqlsh\) lab](#).

▼ Click here for Solution

Run the below command on the terminal.

```
1 exit
```